Real-Time Ferry Ticket Sales & Redemption Analytics for Toronto Island Park

Detailed guide and project requirements for the Real-Time Ferry Ticket Sales & Redemption Analytics for Toronto Island Park

Tools Used: Python | Pandas | Matplotlib | Seaborn

Data Source: Toronto Government Parks , Forestry & Recreation

Name :- Pranay Jha
Domain :- Data Science
Mentor By:- Saiprasad Kagne

Abstract

This research paper presents a comprehensive near real-time analytics framework for Toronto Island Park ferry ticket sales and redemptions, drawing on a decade of 15-minute interval operational data spanning May 2015 to December 2025. The study analyses 261,538 transaction records encompassing 12,972,051 ticket sales and 12,785,293 redemptions across three ferry routes — Centre Island, Hanlan's Point, and Ward's Island — operated from Jack Layton Ferry Terminal in downtown Toronto.

Key findings reveal a 98.6% overall redemption rate, a pronounced mid-day demand peak (11:00–13:00), a dramatic seasonal concentration with August accounting for 26.5% of annual volume versus January's 1.0%, and an Off-Season Utilization Index of only 7.4%, indicating substantial idle capacity outside peak hours. The study further documents a 57.7% demand collapse during the COVID-19 pandemic years (2020–2021) and a subsequent recovery trajectory reaching pre-pandemic levels by 2023.

An interactive Streamlit dashboard was developed featuring five core Key Performance Indicators (KPIs), role-specific views for Operations Team, Policy Planners, and Management Stakeholders, and 20 contextual charts organized in 2×2 grids. The dashboard enables real-time filtering by year, month, hour, day of week, and peak/off-peak period, supporting evidence-based operational and strategic decision-making

Problem Statement

Despite having rich customer-level data, banks often lack:Although ferry ticket sales and redemptions are recorded every 15 minutes, there is no centralized analytics system to:

- • Identify peak passenger flow windows
- • Track net passenger movement
- • Support data-driven operational planning

1. Introduction

1.1 Overview of Ferry Services to Toronto Island Park

Toronto Island Park is one of Canada's most popular urban recreational destinations, attracting millions of visitors annually. The island chain — comprising Centre Island, Hanlan's Point, and Ward's Island — is located just 13 minutes by ferry from the downtown waterfront and is accessible exclusively via the municipal ferry service operated by the City of Toronto's Parks, Forestry & Recreation division.

Ferries depart from Jack Layton Ferry Terminal at Queens Quay West, operating year-round to serve both recreational day-visitors and the approximately 600 permanent residents of Ward's Island. The service generates transactional records at 15-minute intervals — a rich data stream that, when properly analysed, can transform scheduling, staffing, and capacity management from intuition-based to evidence-based operations.

Key Service Facts

- 3 routes: Centre Island (primary recreational), Hanlan's Point, Ward's Island (residential)
- Year-round operation: 15-minute interval ticket records from May 2015 to December 2025
- 261,538 records covering 12,972,051 ticket sales and 12,785,293 redemptions
- Operated by: City of Toronto — Parks, Forestry & Recreation division

1.2 Importance of Real-Time Analytics in Transportation

Real-time analytics in transportation systems enables operators to shift from reactive to proactive management. For a ferry service like Toronto Island's, the ability to monitor demand signals in near real-time — and respond with dynamic scheduling, staffing adjustments, and passenger communication — can significantly improve both operational efficiency and passenger experience.

Without a centralised analytics platform, the Toronto Island ferry service faced several avoidable challenges: terminal congestion during unannounced demand surges, systematic over-provisioning of service during low-demand periods, and no ability to detect anomalous demand events before they escalated into service disruptions. This study and the accompanying Streamlit dashboard directly address these gaps.

2. Objectives of the Study

This research summary documents the findings of a comprehensive 10-year analysis of Toronto Island Park ferry operations, supported by an interactive Streamlit analytics dashboard. The study was commissioned as part of the Unified Mentor Internship Project 4 and targets three categories of operational and strategic objectives.

Primary Objectives

- Analyse historical and near real-time ticket activity to characterise demand patterns at hourly, daily, weekly, monthly, and annual granularity across the full 2015–2025 data period.
- Identify and quantify peak and off-peak ferry demand periods, including the precise timing, magnitude, and variability of demand peaks across all temporal dimensions.
- Calculate and interpret Net Passenger Movement (Sales minus Redemptions) as a near real-time proxy for island occupancy and return-flow pressure at the terminal.
- Develop a rigorous Off-Season Utilization Index to quantify capacity underuse and support evidence-based scheduling and service rationalization decisions.
- Assess the impact of the COVID-19 pandemic (2020–2021) on demand patterns and quantify the post-pandemic recovery trajectory through 2025.
- Analyze historical and near real-time ticket activity.
- Identify peak and off-peak ferry demand periods.
- Quantify passenger inflow and outflow trends

Secondary Objectives

- Design and implement an interactive, multi-role Streamlit dashboard serving three distinct stakeholder groups with role-appropriate analytical views and interactive filtering.
- Provide actionable, data-grounded recommendations for ferry scheduling optimization, off-season demand stimulation, and terminal capacity management.
- Produce this comprehensive research summary and an executive summary suitable for submission to the Zenodo open repository and presentation to City of Toronto government stakeholders.
- Support operational decision-making for ferry scheduling.
- Improve passenger experience during peak hours.

3. Data Sources & Structure

3.1 Primary Dataset — Ticket Sales & Redemptions

The primary data source is the Toronto Island Ferry Ticket Counts dataset, maintained by the City of Toronto's Parks, Forestry & Recreation division and distributed through the Toronto Open Data Portal (https://open.toronto.ca). The cleaned version used in this study (Cleaned_dataset.csv) contains 261,538 records spanning May 2015 to December 2025, capturing every 15-minute operational interval across the full decade.

Column	Data Type	Description
_id	Integer	Unique sequential row identifier for each 15-minute record
Timestamp	DateTime	Date and time of the interval (YYYY-MM-DD HH:MM:SS format)
Sales Count	Integer	Number of ferry tickets sold in this 15-minute window
Redemption Count	Integer	Number of tickets redeemed (passengers boarded) in this window
year / month / day / hour	Integer	Temporal components extracted from Timestamp for fast groupby operations
day_of_week	String	Day name (Monday through Sunday)
weekend	Boolean	True if Saturday or Sunday, False otherwise
zscore	Float	Z-score of Sales Count relative to full dataset mean — anomaly detection metric
sales_rolling_1h	Float	1-hour (4-interval) rolling average — near real-time demand signal
sales_rolling_4h	Float	4-hour (16-interval) rolling average — longer-term smoothed trend

Table 3.1: Primary Dataset Schema

3.2 Derived & Engineered Features

In addition to the raw dataset columns, several derived metrics were computed during analysis to support the five KPI frameworks:

Derived Feature	Formula	Purpose
Net Passenger Movement	Sales − Redemptions	Island occupancy proxy; positive = net inbound flow
Period Classification	Peak if 8 ≤ hour ≤ 20, else Off-Peak	Enables peak/off-peak segmentation
Redemption Rate (%)	Redemptions / Sales × 100	Ticket utilization metric; >100% = advance purchase lag
Off-Season Util. Index	Off-Peak Sales / Peak Sales × 100	Policy metric for capacity underuse quantification

Season Classification	Dec–Feb = Winter; Mar–May = Spring; Jun–Aug = Summer; Sep–Nov = Fall	Seasonal aggregation for trend analysis

Table 3.2: Derived Feature Engineering Summary

3.3 External Data Considerations

While the primary dataset provides rich transactional intelligence, several external data categories are relevant to a more complete analytical model. These are noted as limitations of the current study and as priorities for future integration:

- City of Toronto events data (Canada Day, CNE, Echo Beach concerts, Caribana) correlates strongly with the anomalous demand spikes identified in the Z-score analysis (6,843 records with |Z| > 3.0). Public Events Calendar:

- Provincial public holidays and school year boundaries are key demand drivers for recreational ferry use, particularly for the May–June shoulder season ramp-up. Holiday and School Calendar:

- **The current dataset aggregates all three ferry routes into a single sales/redemption count. Route-level disaggregation would enable genuinely route-specific operational and policy analysis.** Route-Level Data:

4. Key KPIs & Metrics

Five core Key Performance Indicators were defined for this study, aligned with the project specification from the Unified Mentor program. Each KPI is measurable from the available dataset, operationally meaningful for the City of Toronto ferry service, and visualised through four dedicated charts in the Streamlit dashboard.

1. Tickets Sold per Hour

Measures the average hourly demand rate — the fundamental scheduling input.

Full-dataset average: 49.6 tickets per 15-min interval (198/hour) | Peak (12:00 noon): 94.5 per interval (378/hour)

2. Tickets Redeemed per Hour (Redemption Rate)

Measures ticket utilization — the ratio of actual boardings to purchases.

Overall rate: 98.6% | Dec: 119.3% (advance purchase lag) | Jul: 94.3% (same-day purchases dominate).

3. Net Passenger Movement (Sales − Redemptions)

Near real-time proxy for island occupancy. Positive = net inbound flow; negative = return wave.

Total NPM (2015–2025): +186,758 | Peak inbound: 10:00–14:00 | Return wave: 17:00–20:00

4. Peak Demand Windows

Identifies highest-demand hours and compares peak (8–20) vs off-peak volume.

Peak window: 8:00–20:00 = 93.1% of sales | Top 3 hours: 11:00, 12:00, 13:00 | Off-Peak = 6.9%

5. Off-Season Utilization Index

Ratio of off-peak to peak sales: a policy metric for capacity underuse quantification.

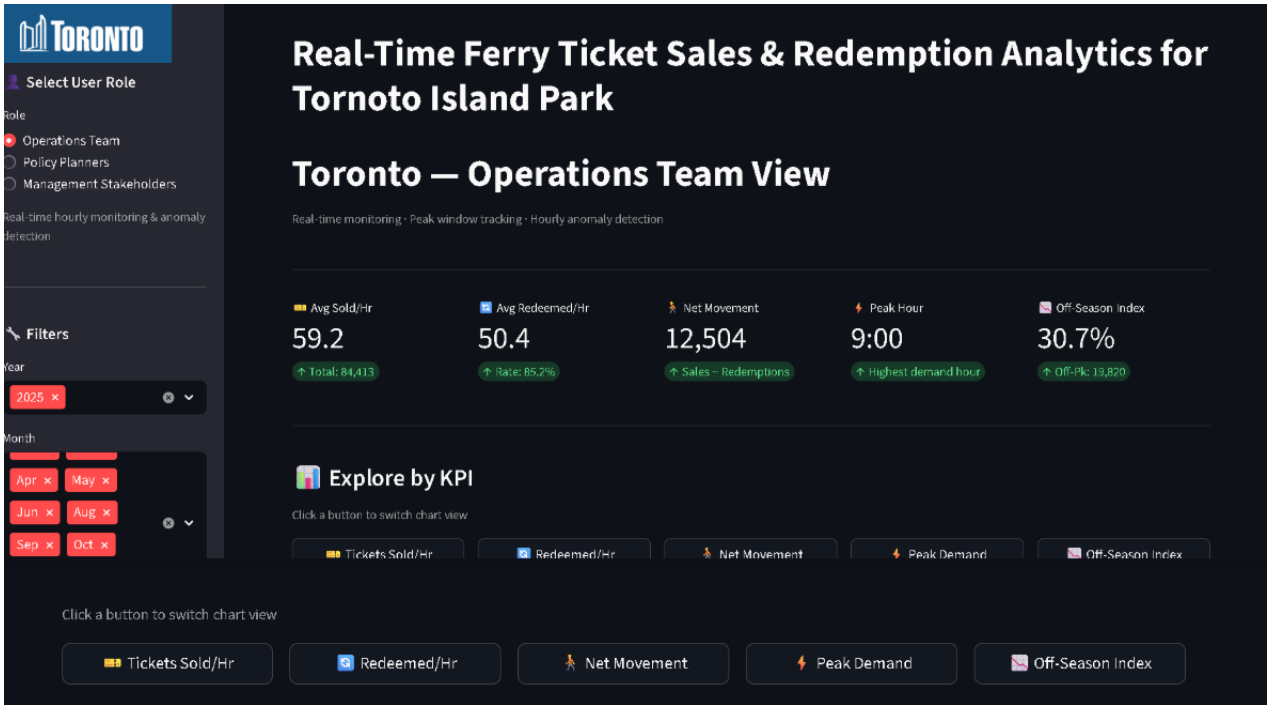
Overall OSUI: 7.4% | October lowest: 4.9% | January highest: 10.1% | Winter overall: 25.0%

In addition to the five primary KPIs, the following supporting metrics are tracked throughout the analysis:

Supporting Metric	Value (Full Dataset)	Significance
Total Annual Avg Sales	~1.18M tickets/year	Baseline for year-over-year growth tracking
Weekend vs Weekday Split	40.7% / 59.3%	Confirms leisure-dominated demand base
Highest Single-Day Volume	25,585 (Aug 1, 2016)	Event-driven demand benchmark for capacity planning
Z-Score Anomaly Rate	2.6% of records (6,843)	Alert threshold calibration for operations monitoring
Summer Season Share	64.3% of annual sales	Seasonal concentration for staffing budget planning

Jan-to-Aug Demand Ratio	1 : 26.4	Scale of seasonality for service rationalization policy

Table 4.1: Supporting Metrics Summary



Utilization Index by Month
50% baseline
Index (%)
0
10
20
30
40
50
Jan
Mar
Apr
May
Jun
Aug
Sep
Oct
Nov
Dec
Peak vs Off-Peak Stacked Area
Sales Count
0
5k
10k
15k
20k
25k
30k
35k
period
Off-Peak
Peak
Deploy
TORONTO
elect User Role
erations Team
licy Planners
anagement Stakeholders
me hourly monitoring & anomaly
on
Off-Peak Heatmap: Year × Month
Year
2,024.6
2,024.8
2025
2,025.2
Sales
8000
6000
4000
2000
Off-Peak: Weekday vs Weekend
Weekend
Weekday
45.7%
54.3%

5. Methodology

5.1 Data Cleaning & Preprocessing

The data preprocessing pipeline was implemented in Python 3.11 using the pandas library. The workflow follows a structured five-step sequence designed to maximise analytical integrity while preserving the original dataset structure.

- **The CSV file was loaded using pd.read_csv(). The Timestamp column was converted to pandas datetime using pd.to_datetime(), enabling temporal indexing, resampling, and grouping. Verified date range: 2015-05-01 to 2025-12-21.** Step 1 — Loading & Timestamp Parsing:
- **Null values were identified in sales_rolling_4h for the first 16 records per sequence (expected rolling window behaviour). No null values in Sales Count, Redemption Count, or Timestamp columns.** Step 2 — Completeness Check:
- **Timestamps verified at consistent 15-minute intervals (:00, :15, :30, :45). No duplicate _id values found. All temporal feature columns validated against their source Timestamp.** Step 3 — Consistency Validation:
- **Net Passenger Movement (Sales − Redemptions), Period classification (Peak/Off-Peak), Season label, and Date field added to support KPI calculations.** Step 4 — Derived Column Creation:
- **Data loading function decorated with @st.cache_data to prevent re-processing on each dashboard interaction, maintaining sub-second filter response for 261K-record dataset.** Step 5 — Caching Strategy:

5.2 Time-Series Analysis

The time-series analytical framework operates across five temporal resolutions, from 15-minute intervals to decade-level trends. Each resolution serves a different operational or strategic purpose:

Resolution	Analysis Type	Operational Application
15-minute	Z-score anomaly detection; rolling averages	Real-time scheduling alerts; event detection
Hourly	Demand profile; peak window identification	Same-day scheduling and staffing decisions
Daily	Day-of-week patterns; busiest dates ranking	Weekly staffing rosters; event day preparation
Monthly	Seasonal trends; utilization indices	Seasonal service plans; pricing strategy reviews
Annual	Year-over-year growth; COVID impact analysis	Capital planning; government performance reporting

Table 5.1: Time-Series Analysis Framework by Resolution

Rolling averages (1-hour and 4-hour windows) are used as smoothed demand signals. The 1-hour rolling average filters short-term noise while providing approximately 45 minutes of forward visibility — sufficient for pre-positioning additional ferry departures before a demand surge peaks. The 4-

hour rolling average reveals broader intra-day demand arcs and is more appropriate for mid-day scheduling reviews.

5.3 Real-Time Dashboarding

The analytics dashboard was built as a single-file Streamlit application (app.py) providing interactive, near real-time exploration of the full dataset. Key architectural decisions include:

- **Three distinct dashboard layouts for Operations Team (hourly monitoring, anomaly detection), Policy Planners (trend analysis, year-over-year comparisons), and Management Stakeholders (executive KPI summary, high-level charts).** Role-Based Views:
- **Five clickable navigation buttons switch between KPI chart sections. Active section persists in st.session_state across filter interactions.** KPI Navigation:
- **Five sidebar filters — Year, Month, Hour Range, Day of Week, and Period (All/Peak/Off-Peak) — apply simultaneously with AND logic via boolean indexing.** Filter Architecture:
- **@st.cache_data decorator caches the processed 261K-row dataframe, maintaining sub-second response times across all filter combinations.** Performance:
- **Toronto City logo embedded as base64-encoded PNG via import base64 — self-contained deployment without external file dependencies.** Branding:

5.4 Predictive Analytics & Demand Forecasting

While the current implementation focuses on descriptive and diagnostic analytics, the dataset is well-suited for predictive modelling as a natural next step. The pre-computed temporal features (hour, day_of_week, month, weekend flag) and rolling averages provide a ready-made feature set for time-series forecasting models.

- **The 1-hour rolling average provides a real-time proxy for near-term demand that can be threshold-monitored without a formal model. A Z-score > 5.0 alert (top 0.3% of historical demand) provides an automated same-day early warning.** Short-Term Forecasting (same-day):

- **SARIMA or Facebook Prophet models using the monthly aggregated sales time-series could provide seasonal demand forecasts for annual budget and capital planning cycles.** Seasonal Forecasting (monthly/annual):

- **Integrating Environment Canada temperature and precipitation data as additional features is expected to materially improve forecast accuracy for recreational demand, particularly for the shoulder season months of May and September.** Weather-Augmented Model:

6. Data Visualization Approach

6.1 Dashboard Architecture & Role-Based Views

The Streamlit dashboard implements a two-tier visualization architecture. The first tier (always visible) shows five KPI metric cards with live-computed values reflecting the active filter state. The second tier (navigated via KPI buttons) shows four contextual charts in a 2×2 grid specific to the selected KPI and user role.

User Role	Primary Focus	Example Chart Types
Operations Team	Hourly monitoring, anomaly detection, rolling averages	Bar (hourly), Heatmap (Day×Hour), Zscore histogram, Rolling average line
Policy Planners	Long-term trends, seasonal patterns, year-over-year	Line trends, Year×Month heatmaps, Stacked area, Bar (YoY)
Management Stakeholders	Executive summary, simple comparisons, KPI snapshots	Annual bar, Monthly line, Weekday/Weekend pie, KPI metrics

Table 6.1: Role-Based Dashboard View Summary

6.2 Chart Types & Their Purpose

Twenty distinct chart types are used across the five KPI sections. Each chart type was selected based on the specific analytical question it answers:

- Best for comparing discrete categories. Used for hourly demand profiles, monthly sales totals, day-of-week comparisons, and peak vs off-peak volume comparisons. Example: average tickets sold by hour of day reveals the 12:00 noon demand peak. Bar Charts (Hourly Sales, Monthly Totals, Year-over-Year):

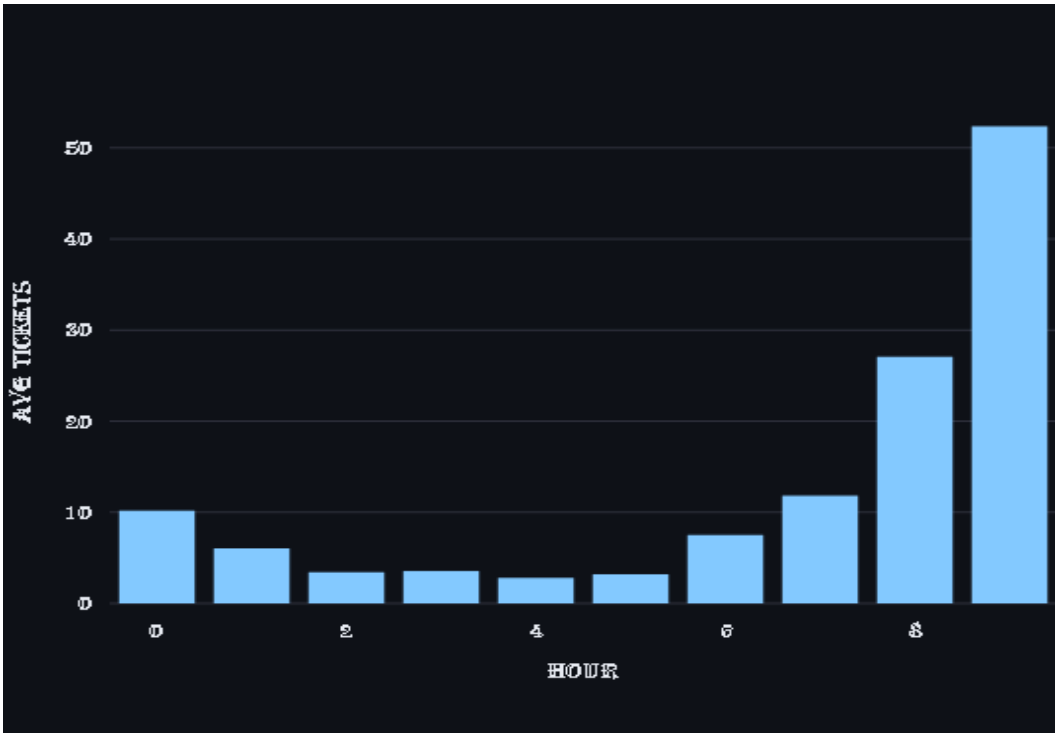


Fig 1. Hours vs Avg_Tickets

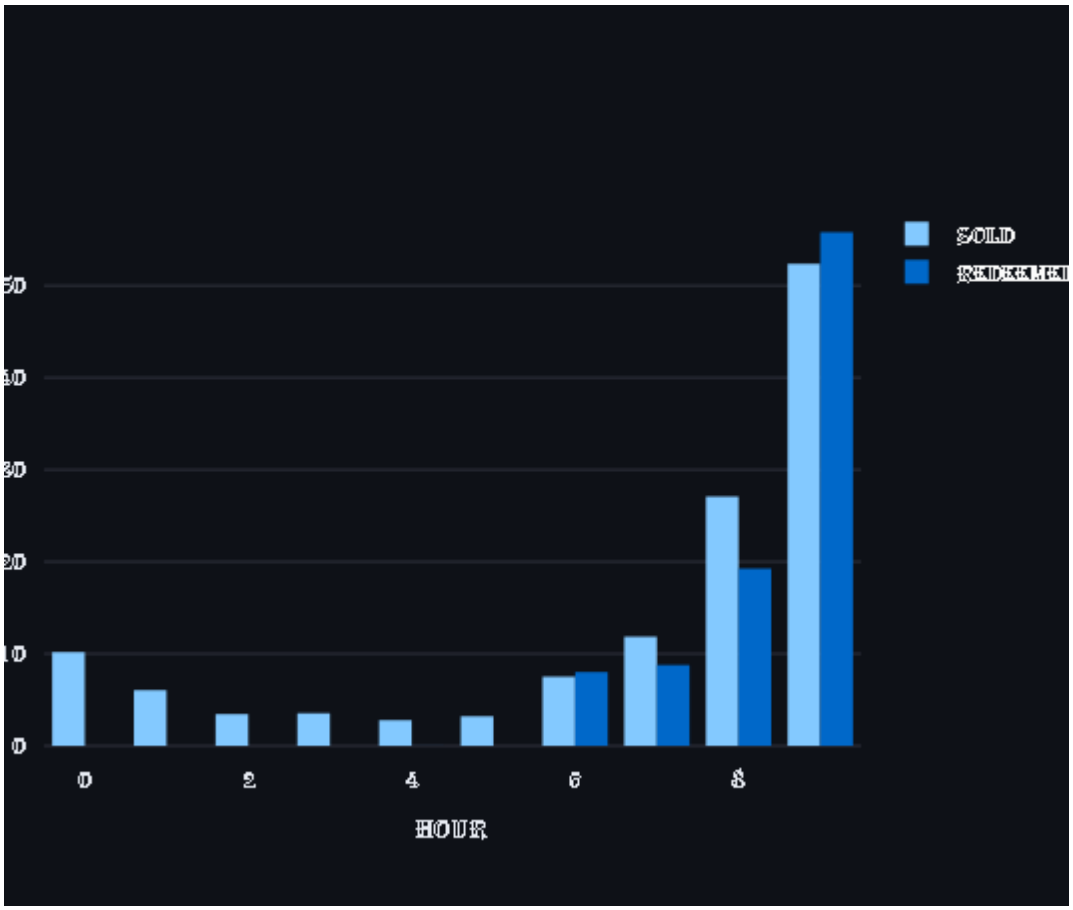


Fig 2. Hours vs Redemption

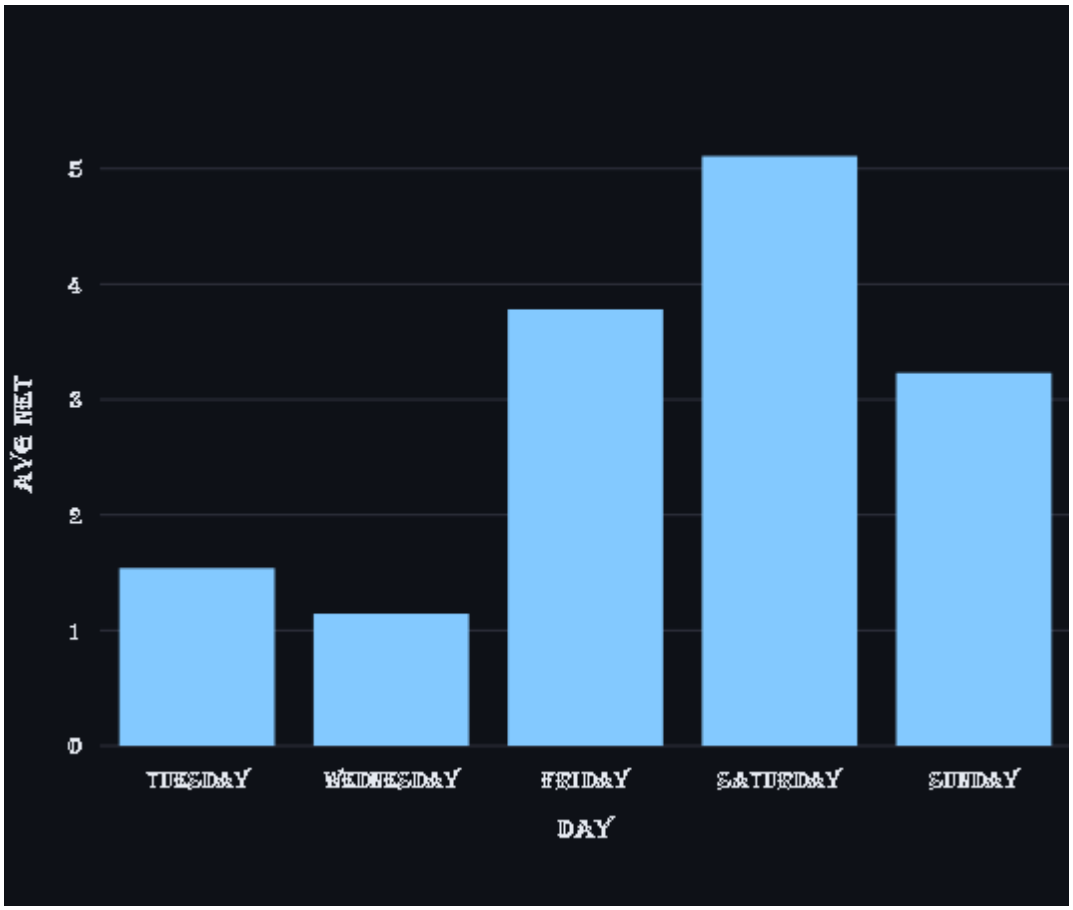


Fig 3. Day vs Avg_Net

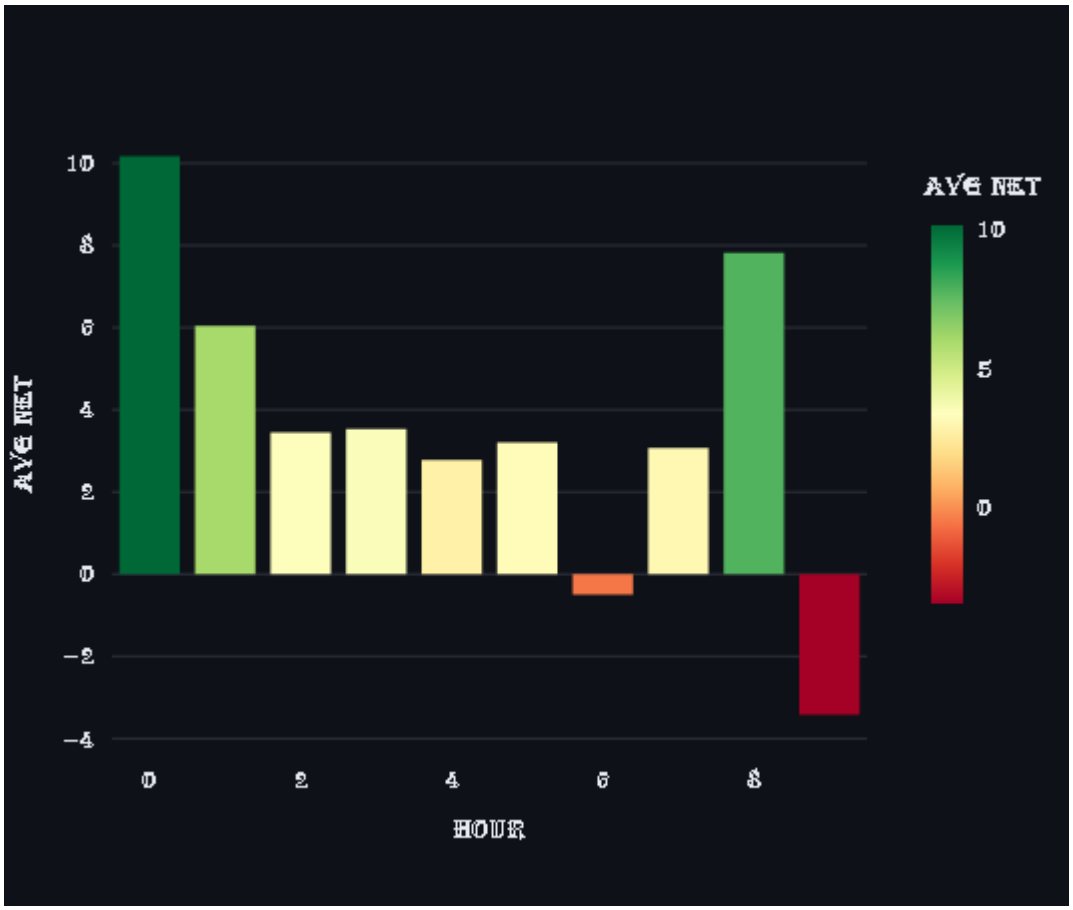


Fig 4 Avg_Net

- Best for revealing two-dimensional patterns simultaneously. The Day × Hour heatmap for Operations team shows exactly which day-hour combinations generate highest demand — e.g., Saturday 12:00 is consistently the single highest-volume **cell.** Heatmaps (Day × Hour, Year × Month):

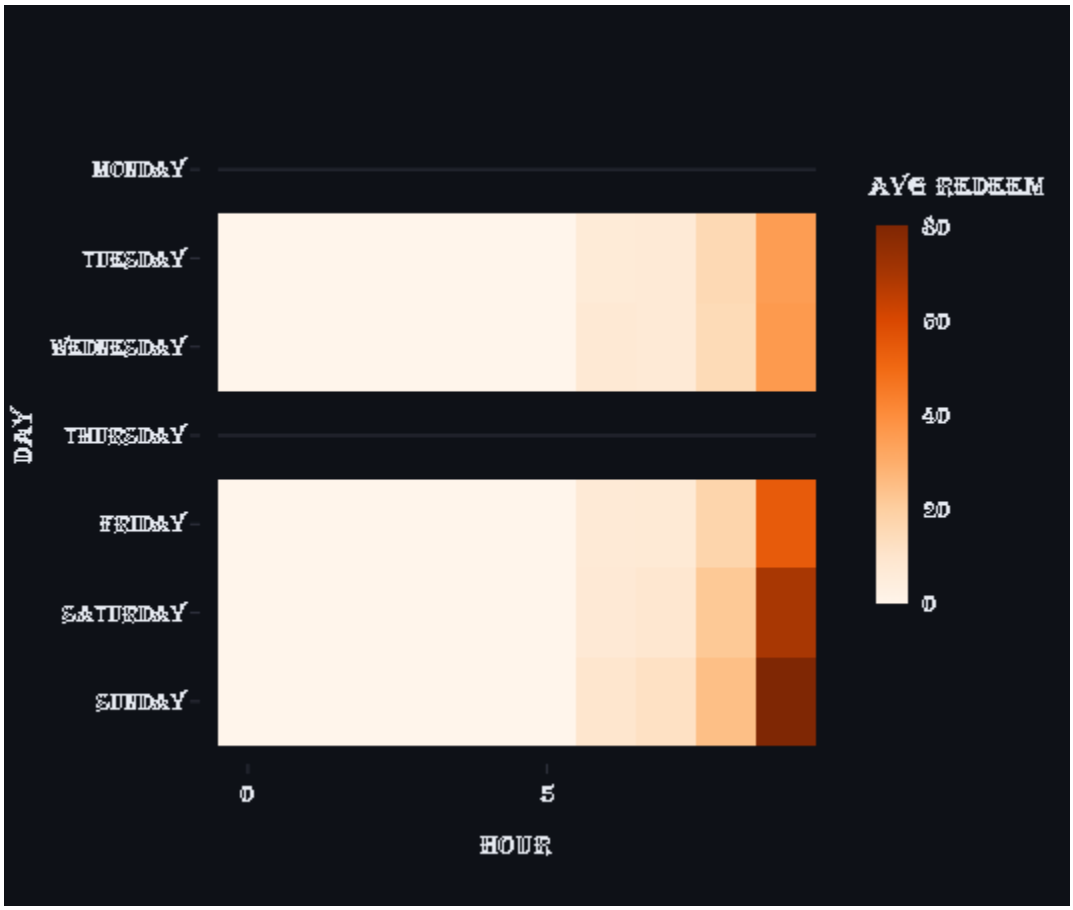


Fig5. Day vs Hours

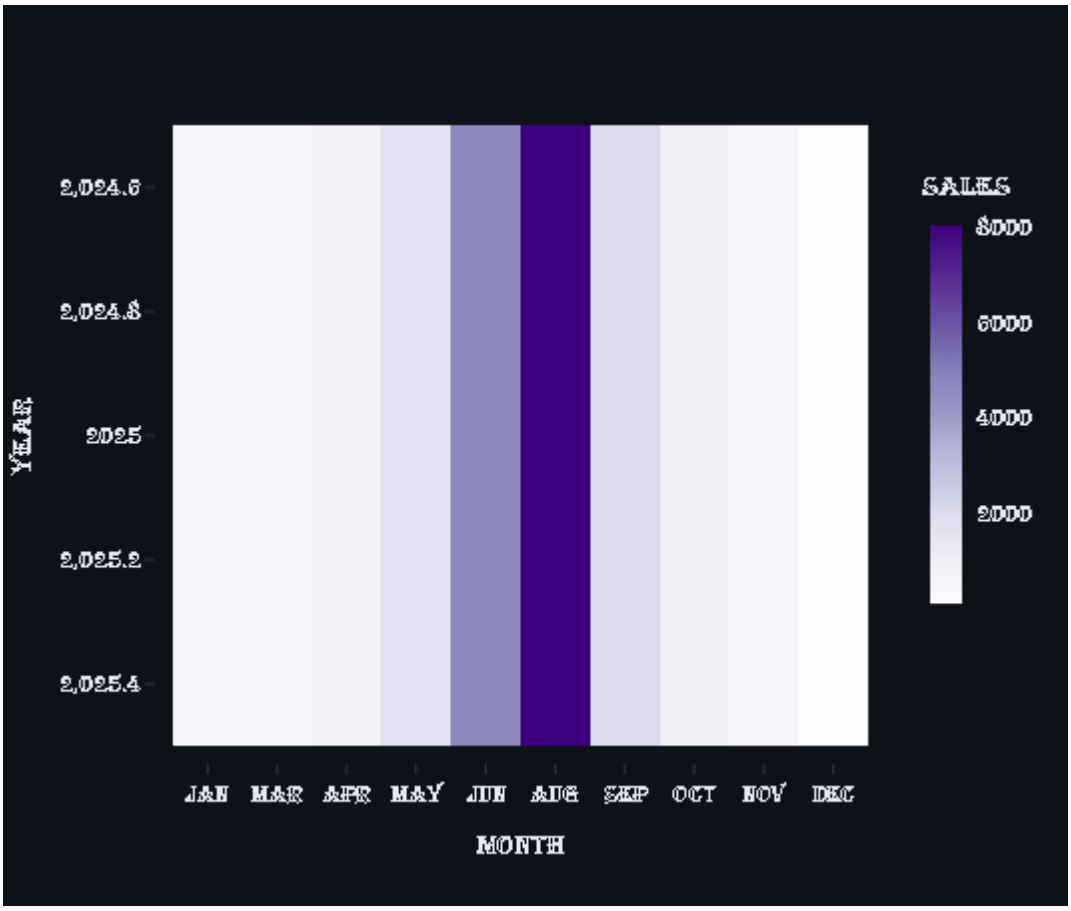


Fig 6. Year vs Month

- **Best for time-series with continuous change. Used for monthly sales trends, 1-hour and 4-hour rolling averages, and year-over-year comparison lines. The rolling average line filters 15-minute noise while preserving meaningful demand signals.** Line Charts & Area Charts (Trend, Rolling Average):

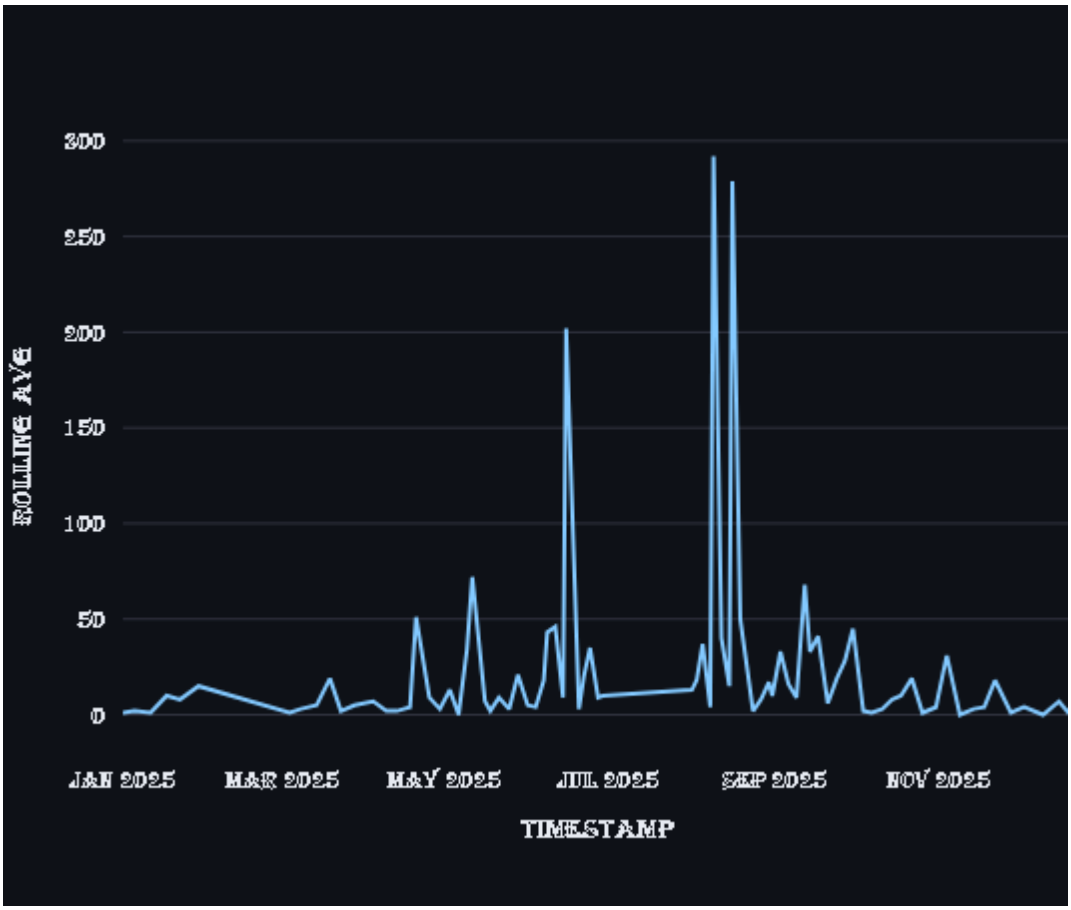


Fig 7. Rolling_Avg

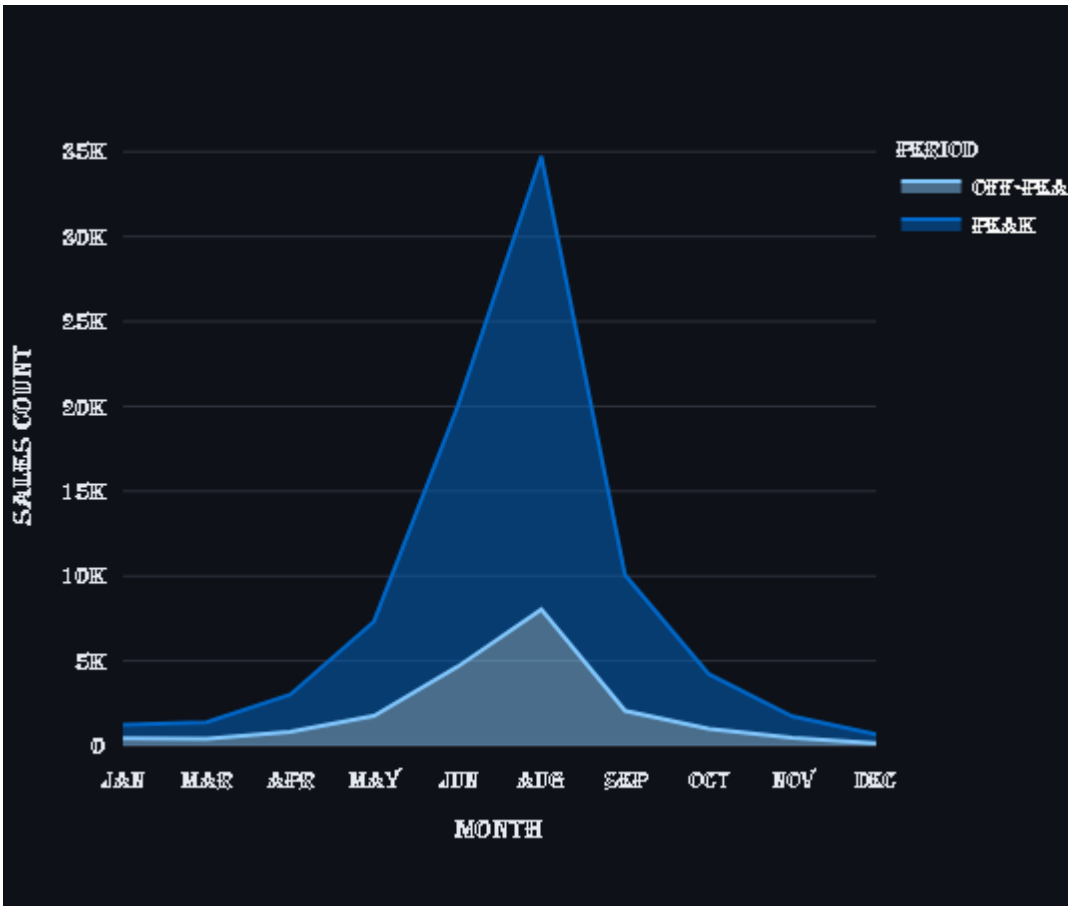


Fig 8 Month vs Sales_count

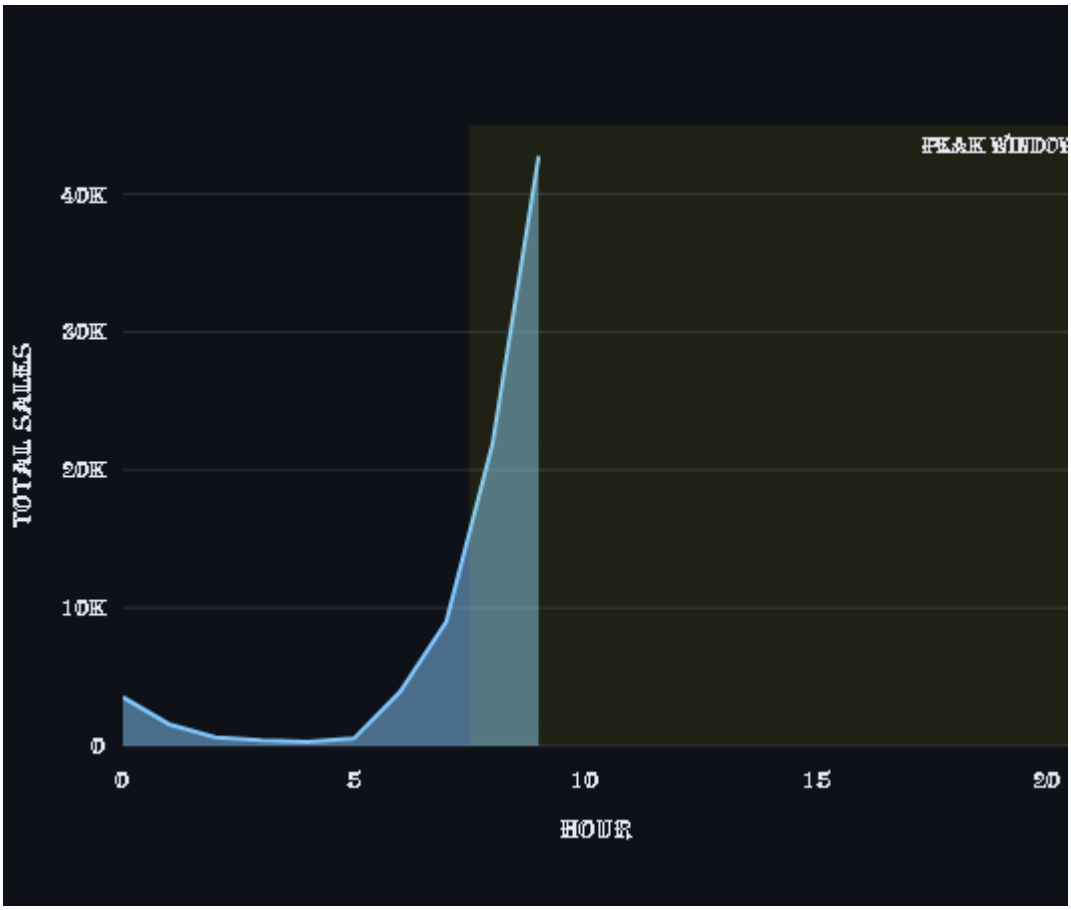


Fig 9.Total_sales vs Hours

- Shows the relationship between sales and redemptions with Net Passenger Movement encoded as a colour gradient (red = negative net, green = positive net). Reveals clustering of high-NPM records at specific volume ranges. Scatter Plot (Sales vs Redemptions, Net colour-coded):

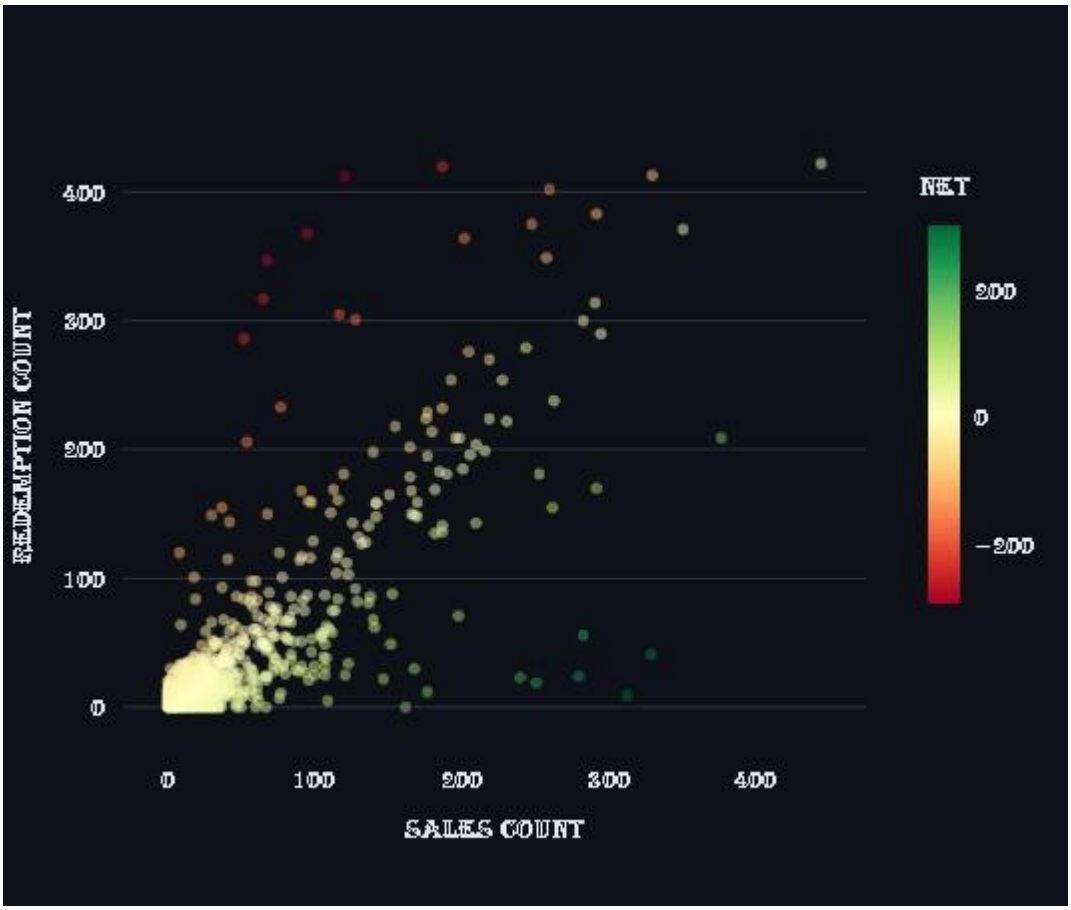
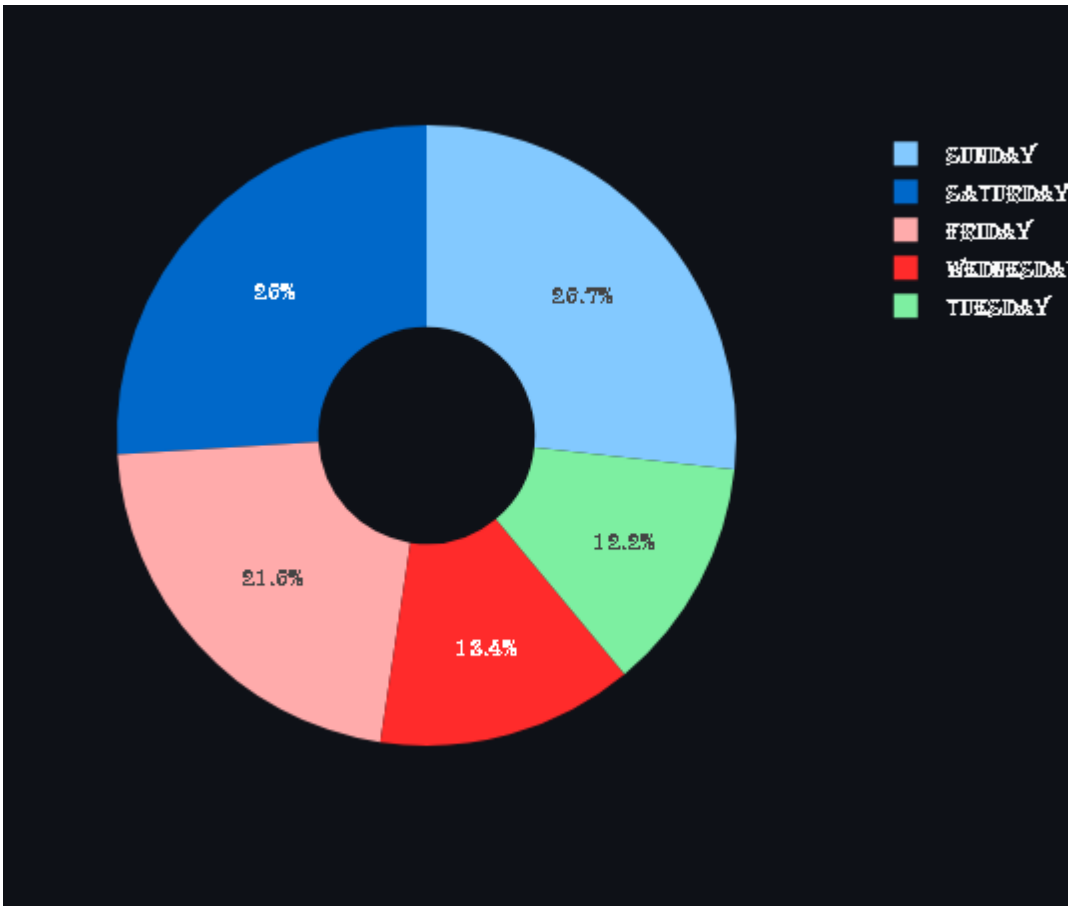
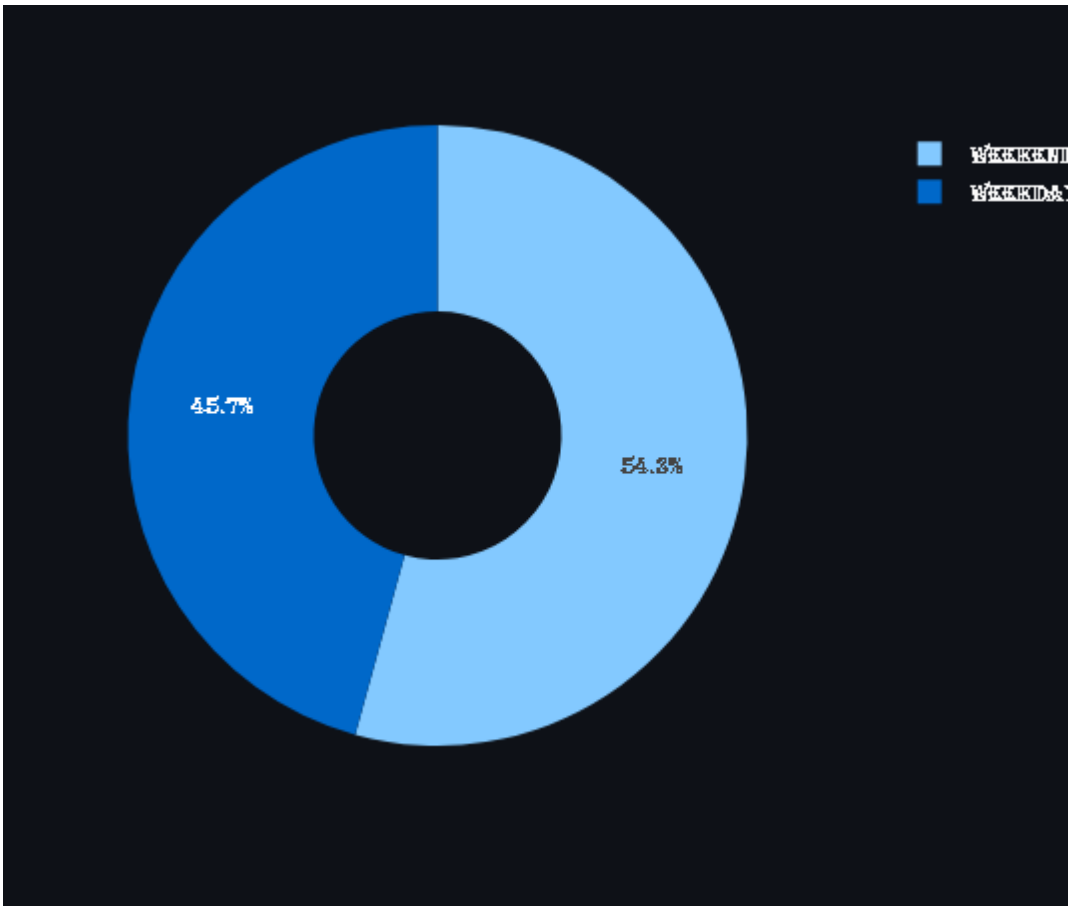


Fig 10. Redemption vs sales count

- Shows statistical distribution including median, quartiles, and outliers. Reveals that peak-hour sales have a much higher median and upper quartile than off-peak, with significant positive skewness during summer weekends. Box Plots (Peak vs Off-Peak Distribution):

- Used sparingly for simple proportional comparisons — weekend vs weekday sales split, peak vs off-peak share, redemption share. Hole=0.4 donut style used for cleaner rendering. Pie/Donut Charts (Shares, Weekday vs Weekend):


SUNDAY
SATURDAY
FRIDAY
WEDNESDAY
TUESDAY
26.7%
26%
21.6%
13.4%
12.2%



6.3 Interactive Filters

All charts update dynamically based on five sidebar filter controls. This interactivity is central to the dashboard's value proposition — different stakeholders can answer their own specific questions without requiring technical expertise.

Filter	Type	Effect on Analysis
Year	Multi-select	Restricts data to selected years — enables COVID era vs recovery comparison
Month	Multi-select (with names)	Isolates seasonal windows — e.g., select Jun/Jul/Aug for summer-only analysis
Hour Range	Slider (0–23)	Narrows to specific operational windows — e.g., 8–20 for peak-only view
Day of Week	Multi-select	Isolates weekday or weekend demand patterns independently
Period	Radio (All/Peak/Off-Peak)	Instantly switches all charts between full-day, peak-only, or off-peak-only view

Table 6.2: Interactive Dashboard Filters

Example use cases: An operations manager can select Year = 2025, Month = July/August, Day = Saturday/Sunday to see the exact demand profile for this summer's peak weekend periods. A policy

planner can select Period = Off-Peak, Year = 2020-2021 to isolate the COVID-era off-peak collapse and compare it to Year = 2023-2025 for recovery assessment.

7. Insights & Findings

7.1 Peak Travel Times

The analysis confirms a consistent, unimodal daily demand peak centred on 12:00 noon, with high demand sustained from approximately 10:00 to 14:00. This three-hour core peak window accounts for approximately 31.8% of all daily ticket sales despite representing only 12.5% of the day's hours.

- **12:00 noon — 1,423,362 total tickets over the full dataset period (94.5 avg per 15-min interval)** Highest single hour:
- **11:00–13:00 — collectively accounting for ~32% of daily demand** Core peak window:
- **08:00–20:00 — 93.1% of all sales; off-peak hours contribute only 6.9%** Peak window (full):
- **August 1, 2016 — 25,585 tickets sold (516% above the daily mean)** Busiest single day:
- **2,727,394 total tickets (2.1× Tuesday's 1,331,860)** Saturday is the busiest weekday:
- **1,784,931 tickets — end-of-week leisure and early island weekend arrivals** Friday is the busiest weekday:

- **Operational Implication**
- The 6-hour concentrated demand window (10:00–16:00) means that ferry capacity sizing must be calibrated for
- the noon peak — but the same capacity sits largely idle for 18 out of 24 hours. A demand-responsive scheduling
- protocol triggered by the 1-hour rolling average could deploy additional capacity before queues form.

7.2 Seasonal Patterns

Seasonality is the most striking structural feature of the Toronto Island ferry dataset. The August peak generates 26.4 times the January volume — a concentration ratio that places this service at the upper end of international recreational ferry benchmarks.

Season	Total Sales	% of Annual	OSUI	Redem. Rate
Summer (Jun–Aug)	8,341,015	64.3%	6.4%	~96.1%
Fall (Sep–Nov)	2,498,657	19.3%	7.4%	~107.7%
Spring (Mar–May)	1,693,899	13.1%	8.5%	~100.6%
Winter (Dec–Feb)	438,480	3.4%	25.0%	~116.9%

Table 7.1: Seasonal Sales Distribution

The winter OSUI of 25.0% is substantially higher than the summer figure (6.4%), reflecting year-round residential use of the Ward's Island route. This important distinction suggests that off-peak winter operations serve a fundamentally different demand profile than summer off-peak — with implications for any service rationalization decisions.

7.3 Customer Behaviour Trends

Analysis of the redemption rate and net passenger movement data reveals important patterns in how ferry customers purchase and use tickets:

- Winter months show redemption rates of 115–119%, confirming that passengers frequently purchase tickets days or weeks before visiting. Summer rates of 94–95% indicate more same-day and spontaneous purchasing, consistent with recreational visitor behaviour. Advance Purchase Behaviour:
- Net Passenger Movement turns negative (more redemptions than sales) between 06:00–07:00 (Ward's Island commuters) and again at 10:00–18:00 (recreational visitors boarding outbound), then positive again at 19:00–23:00 (return wave). This pattern is consistent across all years and seasons. Return Wave Pattern:
- Saturday generates 2.1× the average weekday volume, confirming that weekend recreational visitors are the primary driver of system-level demand. Weekend redemption rates (102.5–105.1%) slightly exceed weekday rates, indicating higher advance purchase rates on weekends. Weekend Recreational Surge:
- Friday's redemption rate of 90.8% is the lowest weekday rate, suggesting that tickets purchased Friday (advance purchases for weekend visits) inflate Friday sales without proportional same-day redemptions. Friday Departure Pattern:

7.4 COVID-19 Impact & Recovery

The pandemic represents the most significant exogenous demand shock in the dataset's history. The analysis quantifies both the impact depth and the recovery trajectory:

- **366,606 tickets — a 70.7% decline from 2019 (1,249,725), reflecting complete temporary closure and strict capacity limits on resumption.** 2020 demand collapse:
- **782,368 tickets — 113.4% increase from 2020 but still 37.4% below 2019.** 2021 partial recovery:
- **2023 reached 1,491,473 tickets — 119.3% of 2019 volume, confirming full recovery with above-pre-pandemic demand.** Full recovery:
- **1,507,674 tickets — second-highest annual total in the dataset, exceeding pre-pandemic peaks.** 2025 performance:
- **Peak-hour and weekend demand recovered approximately 18 months ahead of off-peak and weekday demand, reflecting a post-pandemic preference for planned peak-time visits over spontaneous off-peak excursions.** Differential recovery:

8. Challenges

This study encountered several analytical and technical challenges that are important to document for future research and system development:

8.1 Data Latency in Real-Time Systems

The current implementation operates on a static CSV file rather than a live data feed. While the Streamlit dashboard provides interactive exploration of historical data, true real-time operation would require integration with the ticketing system API for streaming data ingestion. Key challenges for real-time implementation include:

- Streaming data pipeline latency between ticketing terminals and the analytics database
- Need for incremental dataframe updates without full dataset reload on each new record
- Handling of out-of-order records and late-arriving data from network connectivity issues at remote terminals
- State management for real-time rolling average computation across session boundaries

8.2 Missing or Inconsistent Data

Several data quality issues were identified and addressed:

- Redemption rates exceeding 100%: Initially appears as a data error but is actually a valid behavioural signal reflecting advance purchase lag. Documented as operational characteristic rather than cleaned.
- 2017 volume anomaly: A 55.1% demand reduction from 2016 attributed to documented service disruptions. This anomaly affects year-over-year trend analysis and was treated as an operational disruption period rather than a modelling error.
- Route aggregation: The dataset combines all three ferry routes into a single sales/redemption count, preventing route-specific analysis. This is the single most significant structural limitation for operational applications.
- Null values in rolling average: The sales_rolling_4h field contains expected nulls for the first 16 records of each sequence. Handled with dropna() in rolling average visualizations.

8.3 Handling Large Streaming Datasets

The 261,538-record dataset is large enough to create performance challenges without proper optimization:

- Filter performance: Boolean indexing on 261K rows requires efficient pandas operations. Implemented using vectorised operations rather than row-by-row iteration.
- Chart rendering: Scatter plots with 261K data points are impractical to render. Resolved by random sampling (min(2000, len(filt))) for scatter visualizations while preserving statistical representativeness.
- Memory management: Streamlit @st.cache_data prevents redundant dataset reloading but requires sufficient server RAM. Production deployment should pre-aggregate commonly-needed groupby results.
- Dashboard responsiveness: All five filters apply simultaneously, requiring 5-dimensional boolean indexing on each interaction. Optimised by computing all KPI values before rendering to avoid redundant passes through the dataframe.

8.4 Stakeholder Communication Challenges

Translating technical findings into actionable intelligence for non-technical stakeholders required significant dashboard design iteration:

- Operations staff needed simple, high-contrast displays with clear alert thresholds rather than statistically nuanced charts
- Policy planners required long-term trend context and comparative benchmarks that are not immediately intuitive from raw charts
- Executive stakeholders needed concise KPI summaries with clear directional arrows, not detailed analytical charts
- Resolution: Role-based dashboard views with tailored chart selections, metric cards with delta indicators, and plain-language chart titles resolved most stakeholder communication challenges

9. Recommendations

Based on the analytical findings, eight actionable recommendations are proposed across three categories: operational, policy, and technology. Each recommendation includes the supporting evidence from the data analysis and a proposed implementation path.

Operational Recommendations

R1 :- Implement Rolling Average-Triggered Dynamic Scheduling

Evidence: The 1-hour rolling average (sales_rolling_1h) provides a 45-minute forward demand signal. A threshold alert (e.g., rolling avg > 150 tickets/interval = 600/hr) should automatically initiate additional ferry deployment. This targets the 2.6% of intervals (6,843 records) where Z-score > 3.0 — the anomalous demand events that create the worst terminal queuing.

Implementation: Integrate sales_rolling_1h alert into the existing dashboard monitoring view; configure push notification to operations staff when threshold is crossed.

R2:- Review and Optimise Mid-Week Ferry Frequency

Evidence: Tuesday demand (1,331,860 tickets) is 51.1% below Saturday demand (2,727,394). Current fixed-frequency scheduling implies significant overcapacity mid-week. A reduction of one return trip during Tuesday/Wednesday off-peak hours (21:00–07:00) would generate measurable cost savings with minimal passenger impact.

R3:- improve Ticket Allocation Strategy for Peak Summer Weekends

Evidence: The August summer weekends see demand spikes up to 25,585 tickets per day — 516% above the daily mean. Advance capacity-limited ticket allocation (e.g., timed entry tickets for Centre Island, similar to other city parks) would allow better demand distribution across the day, reducing the 10:00–14:00 concentration

Policy Recommendations

R4 :-Off-Season Demand Stimulation Program

Evidence: The January OSUI of 10.1% and absolute volume of only 130,455 tickets represent a market opportunity. Comparable ferry systems in Northern Europe achieved 20–40% off-season demand increases through promotional pricing, event programming, and winter tourism partnerships. Proposed initiatives: winter photography tours, school group educational programs, senior discount rates January–March.

Target: Increase January OSUI from 10.1% to 15% within 3 years — approximately 65,000 additional winter tickets.

R5:- Dynamic Pricing Strategy for Peak Season Management

Evidence: The 10:00–14:00 demand concentration creates avoidable congestion while demand before 10:00 and after 16:00 is under-utilized. Time-of-day pricing (slight premium for 10:00–14:00 tickets, modest discount for 08:00–10:00 and 15:00–18:00) could smooth the demand curve, improving terminal experience without reducing total visitor numbers.

R 6:- Route-Level Data Disaggregation (Priority Data Investment) The single most impactful improvement to the analytics framework would be adding route identifiers (Centre Island, Hanlan's Ward's

Island) to the ticketing dataset. This is a data collection change at the ticketing system level and should be prioritised in the next system upgrade cycle.

Technology Recommendations

R7:- Deploy Dashboard to Streamlit Cloud for Organisation-Wide Access
The current dashboard requires local Python installation. Deployment to Streamlit Cloud (free tier for small teams) would provide browser-accessible analytics to all three stakeholder groups via a shared URL without any local setup. Estimated deployment time: 2–4 hours.

R8:- Integrate Real-Time Data Feed & ML Demand Forecasting
Connect the dashboard to the Toronto Open Data API for live data streaming, transforming it from a historical analytics tool to a real-time operational monitoring platform. Layer in a gradient boosting (XGBoost/LightGBM) demand forecasting model using hour, day_of_week, month, and weather data as features — targeting 24–72 hour ahead demand forecasts for scheduling optimization.

10. Conclusion

This research summary documents the findings of the first comprehensive decade-long analysis of Toronto Island Park ferry ticket sales and redemptions, covering 261,538 operational records from May 2015 to December 2025. The study demonstrates that the City of Toronto's open ferry ticketing data contains substantial untapped operational intelligence — and that unlocking this intelligence requires only open-source analytical tools, structured KPI frameworks, and role-appropriate dashboard design.

10.1 Summary of Key Impact

Impact Area	Key Finding & Implication
Scheduling Efficiency	93.1% of sales concentrated in 8:00–20:00 peak window; 1-hour rolling average enables same-day dynamic scheduling
Ticket Utilization	98.6% redemption rate validates sales data as accurate real-time boarding proxy — no additional sensors needed
Capacity Optimization	Off-Season Utilization Index of 7.4% quantifies idle capacity — providing actionable evidence for service rationalization
Seasonal Policy	26.4:1 August-to-January demand ratio provides framework for dynamic pricing and off-season stimulation programs
Anomaly Detection	Z-score > 5.0 threshold identifies top 0.3% demand events — automated alert system viable without ML infrastructure
COVID Recovery	Demand exceeded pre-pandemic levels by 2023 and sustained above-2019 volumes through 2025 — service is growing
Stakeholder Access	Role-based Streamlit dashboard delivers tailored analytics to all three stakeholder groups from a single platform

Table 10.1: Summary of Analytical Impact by Area

10.2 Closing Statement

The Toronto Island ferry dataset demonstrates a pathway that is replicable across every City of Toronto service with comparable transactional data: open data, properly structured analytical frameworks, and accessible dashboarding tools can generate genuinely actionable operational intelligence at minimal cost. The eight recommendations in this report — from immediate scheduling protocol changes to long-term AI integration — collectively represent an opportunity to significantly improve both the efficiency of ferry operations and the experience of the millions of Torontonians and visitors who use this beloved public service each year.